PNAS RESEARCH ARTICLE | SOCIAL SCIENCES

What Drives NWSL Attendance: Evidence of a Durable, Women's-Specific Sports Culture

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This manuscript was compiled on August 27, 2026

Attendance in the National Women's Soccer League (NWSL) has nearly doubled since 2016, but there is little agreement on the the main drivers of this growth, and therefore little basis for judging whether it will last. This paper asks whether NWSL attendance is explained by the same structural factors that drive attendance in established men's leagues, market size and on-field success, or by factors specific to the league's own fan culture. I fit an OLS model of log(attendance) on 1,005 NWSL home games (2016–2025, 2020 and 2021 excluded for COVID), with standard errors clustered by team and season fixed effects, and fit the same model on 3,433 Major League Soccer (MLS) games so that a women's-soccer pattern can be separated from a soccer pattern. A separate analysis characterizes the league's media footprint using LDA topic modeling on 8,328 news headlines. Market size predicts nothing in either league on a matched specification (NWSL 20.5%, p = 0.458; MLS 6.2%, p = 0.480), and points per game is a modest predictor in both. The Cascadia derby, by contrast, is worth +71.5% (p = 0.008) in the NWSL against +27.3% (p = 0.100) for the identical fixture in MLS. The Cascadia derby, by contrast, is worth $+70.8\%$ ($p < 0.001$) in the NWSL against $+29.3\%$ ($p = 0.095$) for the identical fixture in MLS. Over the same window NWSL attendance grew 59.0% while MLS grew 1.1%. The apparent effect of stadium relocation does not survive controls for league-wide growth. News coverage quintupled after 2018 with almost no change in topic composition, and independent women's-sports channels now out-publish the league's own.

National Women's Soccer League | Sports Attendance | Women's Sports Media

Significance

The growth of the National Women's Soccer League (NWSL) is about much more than soccer in the United States; it signals increased investment in women's sports globally, and the development of a new fan that is shifting media economics. Whether that growth is lasting depends on what is driving it. If the NWSL fills stadiums through big markets and winning teams, then the growth is constrained by factors that no front office can control. I test this by running an attendance model on both the NWSL and Major League Soccer over the same seasons, markets, and often stadiums. Market size predicts nothing in either league, and the league grew 59% while MLS stayed flat.

1. Introduction

The National Women's Soccer League (NWSL) commenced its inaugural season in 2013, after the former women's professional league in the United States folded in 2012. In light of this context, as the league began, it faced significant skepticism from players, fans, and the media as to whether it could survive long-term.

Attendance and expansion fee records are broken regularly. Earlier this season, the attendance record was shattered when 63,004 fans watched Denver Summit FC play the Washington Spirit. The trend seems clear: the NWSL is growing- by some metrics at an exponential rate-and the league is set up for long-term success; however, some are more skeptical.

Settling that argument requires knowing what actually produces attendance, and the literature offers two competing accounts of where a team should locate and why fans show up. The *market-access* account holds that attendance follows reachability. One study documents a suburban penalty in the NWSL specifically, estimating with Tobit regression to handle games censored at stadium capacity, that attendance declines roughly 6.6% for every mile a team's stadium sits from its city center (1). A study on the NFL argued that franchises have a strong economic incentive to separate themselves from businesses to force attendees to remain in the stadium and spend more money (2). This manifests as stadiums in suburban areas surrounded by a moat of parking lots.

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There's an established honeymoon effect across sports leagues, where attendance increases by up to 20 percent within 5 years of a stadium's opening, regardless of accessibility increases (3). Other documented attendance factors in the NWSL include post-World Cup booms, star players, and team quality (4). However, there's an interpretive gap. When studying the NWSL alone, there's no way to parse between a women's soccer effect versus a wider soccer effect. The NWSL and MLS share a sport, several markets, and twelve venues in which a team from each league currently plays. Running the same model on both is the cleanest available way to separate the two.

I test three predictions. First, market size and on-field success are weak predictors of NWSL attendance. Second, if that weakness reflects soccer rather than women's soccer, the same weakness appears in MLS. Third, factors rooted in a club's own investment and fan culture, stadium relocation, proximity to downtown, and established rivalry, outperform the structural ones. A secondary analysis then asks whether the league's media footprint shows the same signature of a durable, women's-specific sports culture. The NWSL simply does not have the legacy broadcast rights that established men's teams have, and so the audience for women's soccer has been built through less traditional avenues than legacy broadcast media.

2. Data

2.1. Soccer Game, Match, Stadium, and Player Data. Match, team, roster, and stadium data come from the American Soccer Analysis (ASA) API via the `itscalledsoccer` Python wrapper, covering both the NWSL and MLS: 7,283 games, 50 teams, and 57 stadiums after cleaning. The original unit of analysis is the individual match, which is then aggregated to team-game and team-season datasets. Many stadiums were missing critical metadata (capacity, location, etc.), so the Wikipedia pages for NWSL and MLS stadiums were scraped to obtain that missing information. Thirty-nine games at stadiums with no recoverable location metadata are dropped. A few hand-built lookup tables were necessary. One for the metropolitant population of each NWSL and MLS market, and downtown coordinates for the NWSL markets. Bay FC plays in San Jose, but is coded for the entire Bay Area, and NY/NJ Gotham FC plays in New Jersey, but they are coded to the full New York/Newark metropolitan area. ASA's data collection for the NWSL begins in 2016, despite the league's 2013 founding, so all MLS data is filtered to the same 2016 window for comparability. This excludes the NWSL's first three seasons from every attendance analysis in this paper.

2.2. News articles. The MediaCloud API supplied 8,328 articles published between 2012 and 2024 mentioning the NWSL or women's soccer. The unit of analysis is the individual article; variables used are title, publish date, and outlet. Filtering occurred in two stages. First, a keyword query at the API, then regex on article titles for the league name, team names and nicknames, and "women's soccer." Coverage is bounded by what MediaCloud's index surfaces and by the keyword filter, which will miss articles that discuss the league without using the search terms in its title.

2.3. YouTube data. The YouTube Data API v3 was used to obtain video count, total views, and subscriber count from the official NWSL YouTube channel and five other women's-sports-specific accounts that have sprung up as the league's popularity has grown. The unit of analysis is the individual channel, aggregated to channel-year. This sample is small, with six samples and selected based on my knowledge of the media landscape, so patterns here should be read as descriptive rather than representative of women's-sports media broadly. Monthly Google Trends search interest for the NWSL, MLS, and the two national teams (2012-2026) were also pulled for comparison.

2.4. Data quality. This project is constrained by sample size. There are only 8 usable seasons for analysis (2016–2025, with 2020 and 2021 both excluded for COVID). There are only 5 recent stadium relocations, and several teams who are secondary-tenants of large venues have semi-unknown capacity constraints because there isted capacity is the full bowl even though upper sections are routinely blocked off.

Table 1. Data used for analysis and what it was used for

Unit	Used by	N	Coverage
Team-game (NWSL)	OLS models M1–M3	1,005 home games	2016–2025, 2020–2021 excluded, 14 teams
Team-game (MLS)	OLS benchmark M4	3,433 home games	2016–2025, 2020–2021 excluded, 30 teams
Team-season	Descriptive tables, relocation analysis	∼85 team-seasons	2016–2025, 2020–2021 excluded
Article	Topic model, peak detection	8,328 articles	2013–2024
Channel-year	Media-ecosystem description	6 channels	2013–2026

3. Methods

3.1. Cleaning and merging.. The ASA and Wikipedia stadium tables were merged and deduplicated. Each team's home stadium was assigned per year by which venue hosted the most games, so a team that moved mid-window carries the correct venue on each side of the move. Team lifespans were derived from active seasons, stadium tenants ranked primary or secondary, and remaining coordinate and capacity gaps geocoded or filled by hand.

3.2. Feature construction. Points per game (3/1/0) is computed per team-season from regular-season results only, since knockout games draw different amounts of demand. Market size is $\log_{10}$ metro population. The post-relocation indicator equals 1 from the season a team moved into its new venue onward. Distance from downtown is the haversine distance in miles from that season's venue to the metro's downtown coordinates.

The rivalry indicator marks the Cascadia derby, Portland Thorns FC against Seattle Reign FC, in either direction. At the game level the indicator marks the 27 actual derbies. This rivalry was chosen because its a standout rivalry within the league, and while there are other budding rivalries such as NY/NJ Gotham FC vs Washington Spirit FC, the Cascadia rivalry is the first strong rivalry to be established in the young league.

3.3. Regression. All models are OLS on log(attendance):

$$\log(\text{attendance}_i) = \beta_0 + \beta' x_i + \gamma_{s(i)} + \varepsilon_i, \qquad [1]$$

where x_i holds the five predictors and $\gamma_{s(i)}$ is a season fixed effect. Models are fit on the full sample rather than a train/test split, because the goal is inference on individual coefficients, in particular whether distance from downtown reproduces the 6.6%-per-mile estimate in (1), and that requires standard errors, not a held-out R^2.

Standard errors are clustered by home team because, although there are many more than 14 rows fed into the model, each of the 14 teams has little variation. All of the recent relocations examined occurred between 2021 and 2024, which is also when the league-wide attendance increased sharply, so the effects of relocation are easily confounded with the overall league growth. M1 omits season effects, M2 includes them, and M3 and M4 fit a matched specification (points per game, market size, rivalry, plus season effects) to the NWSL and MLS respectively, dropping distance and relocation because that data is only available for the NWSL.

3.4. Rivalry and away-draw lift.. Alongside the regression, a descriptive lift measures each visiting team's attendance against the host's own same-season baseline rather than the host's career average, which would over-credit visitors who happened to play a fast-growing host early. Folded teams are eliminated and matchup-level comparisons are restricted to pairings with at least eight meetings to avoid bias from the newest expansion teams, which retains 73 of 231 possible matchups.

3.5. Peak detection and topic modeling. `scipy.find_peaks` flags spikes in the daily and monthly article-count series with a prominence score for each. Article titles were tokenized and stemmed with domain-specific stopwords (team names and nicknames, generic league and game words, headline filler). An LDA model was estimated in `gensim` with four topics, ten passes, and automatic α, on a dictionary filtered from 5,076 to 101 terms by dropping terms in under 1% or over 90% of documents. Four topics were chosen because five produced overlapping topics and three left them vague. Each article was tagged with its highest-probability topic; a total of 8,318 articles were successfully tagged.

4. Results

4.1. Growth within the league and compared to the MLS. The NWSL average attendance rose from 5,711 in 2016 to 10,243 in 2025 across the full league, currently 10,944 through the partial 2026 season. The most significant year-over-year jump arrives in 2023, post-COVID, when the league average jumped from 7,170 to 10,061.

Figure 1 shows NWSL growth in comparison to the MLS. On the sample, the NWSL grew 59.0% between 2016 and 2025, from 6,658 to 10,588 fans per game, while MLS moved from 21,692 to 21,926, a change of 1.1%. MLS remains roughly twice the NWSL's size in absolute terms, but it is not growing. Whatever is lifting NWSL attendance is not necessarily lifting American professional soccer as a whole. An important caveat is that attendance does not just reflect demand; it also can reflect a capacity constraint, and since the MLS has much higher attendance, it is likely subject to more capacity constraints. I am comparing leagues in very different stages of maturity.

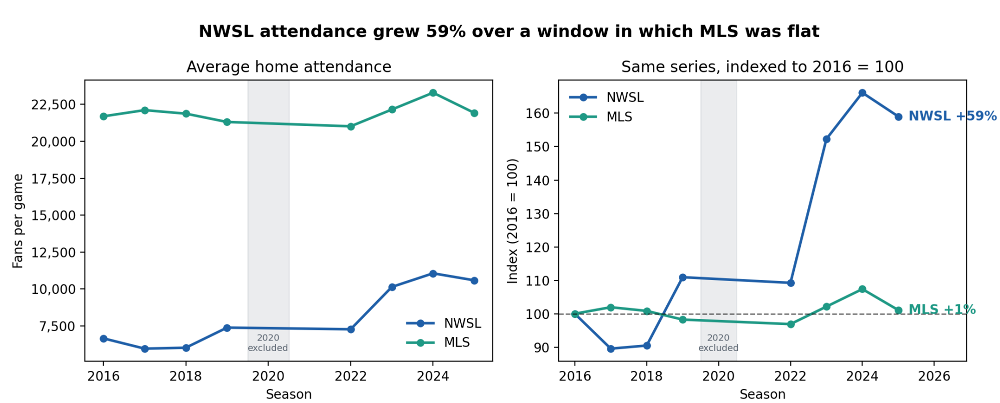

Fig. 1. Average home attendance by season for both leagues on the same panel definition (regular season, active franchises, 2016–2025, 2020 excluded). Left: raw levels, showing MLS roughly twice the NWSL's size. Right: the same series indexed to 2016 = 100, to make a fairer comparison.

While the league as a whole is growing, it is also stratifying. Portland Thorns FC average 17,101 fans across their history while Chicago Stars FC average 4,657 and Gotham average 4,864. Kansas City Current sell out their purpose-built stadium at 100% of capacity, while Seattle Reign fill 18.8% of an oversized shared bowl. Washington Spirit posted the largest 2025 gain in the league at +26.6%.

4.2. Structural predictors are weak in both leagues. Points per game correlates with same-season attendance at $r = 0.240$ across 85 NWSL team-seasons and $r = 0.171$ across 261 MLS team-seasons (2020 and 2021 excluded). Market size correlates at $r = 0.047$ across 16 NWSL markets and $r = -0.192$ across 30 MLS markets. In the MLS, the New York Red Bulls and New York City FC, in by far the largest metro in either league, draw fewer fans than Nashville, Charlotte, or Cincinnati. Similarly, NY/NJ Gotham FC consistently draws some of the lowest attendance in the league, despite representing the largest metro area and consistently performing well on the pitch.

The regression confirms this. Table 2 reports all four models. Market size is indistinguishable from zero in every specification and in both leagues, and its direction is consistently negative (12.4%, p = 0.699 in the NWSL; 6.2%, p = 0.480 in MLS). The lack of significance across both leagues suggests this is a soccer-specific trend, not a women's-specific one.

In the matched specification, points per game is worth +32.4% per additional point per game in the NWSL ($p = 0.018$) and +18.4% in MLS ($p = 0.005$). Winning modestly helps fill stands in both leagues. It is not, however, what distinguishes them, and it is not large enough to account for a 59% rise over a decade in which NWSL team quality did not systematically improve.

4.3. The derby premium is large in the NWSL.. The Cascadia derby is the largest and most precisely estimated effect in the model: +57.5% in the preferred specification (95% CI $[+13.2\%, +119.1\%]$, $p = 0.007$), and +71.5% in the matched specification. Figure 2 ranks all five predictors on the percentage-change scale.

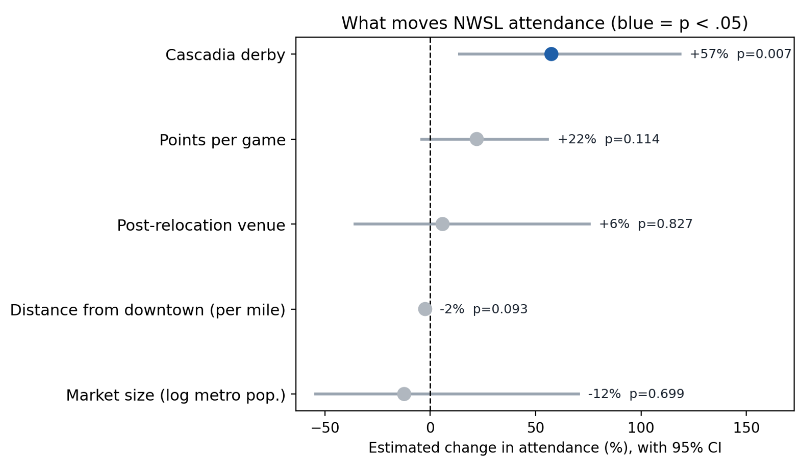

Fig. 2. Estimated percentage change in attendance for each predictor, with 95% confidence intervals, from the preferred NWSL specification (M2: game level, season fixed effects, standard errors clustered by home team). Blue marks $p < .05$.

Table 2. OLS estimates of log(attendance). Cells report the estimated percentage change per unit, with p-values from standard errors clustered by home team. M2 is the preferred NWSL specification; M3 and M4 are a matched specification allowing a direct NWSL/MLS comparison.

	NWSL			MLS
Predictor	M1 (no season FE)	M2 (preferred)	M3 (matched)	M4 (matched)
Cascadia derby	+41.2% (0.096)	+57.5% (0.007)	+71.5% (0.008)	+27.3% (0.100)
Points per game	+12.9% (0.454)	+22.0% (0.114)	+32.4% (0.018)	+18.4% (0.005)
Market size (log metro pop.)	−18.0% (0.587)	−12.4% (0.699)	−20.5% (0.458)	−6.2% (0.480)
Post-relocation venue	+26.8% (0.347)	+5.8% (0.827)	—	—
Distance from downtown (per mile)	−2.9% (0.053)	−2.4% (0.093)	—	—
Season fixed effects	No	Yes	Yes	Yes
N (games)	1,005	1,005	1,005	3,433
R^2	0.213	0.310	0.256	0.049

The MLS's strongest rivalry is also their Cascadia rivalry between the Portland Timbers, Seattle Sounders, and Vancouver Whitecaps, contested over 61 games in the same window, making for a comparable analysis. The identical fixture in the men's league is worth +27.3% and does not reach conventional significance ($p = 0.100$). Comparing the two leagues on the matched specification, the derby premium in the women's league is roughly 2.6 times its men's-league counterpart (+71.5% against +27.3%). NWSL derbies average 14,148 fans against 11,870 for Portland and Seattle's other home games, a 19.2% raw premium, while the MLS Cascadia Cup games average 29,049 against 26,901, an 8.0% premium.

Matchup-level lifts corroborate this. A Portland visit to Seattle draws 3,356 more fans than Seattle's own same-season baseline across 14 meetings, the largest of the 60 qualifying matchups; Washington at Gotham is worth $+2,572$ across 14 meetings, and San Diego at Portland $+1,969$ across nine.

4.4. Distance replicates directionally, but at a third the published magnitude. Distance from downtown is negative in every specification, at -2.9% per mile without season effects ($p = 0.053$) and -2.4% per mile with them ($p = 0.093$). The direction reproduces Chahardovali et al. (1) and favors the market-access account over the enclosure account (2), but the magnitude is roughly a third of the published 6.6% per mile, and the confidence interval in the preferred specification includes zero.

4.5. The stadium effect cannot be separated from league growth. Every one of the five relocations coincided with a large attendance gain: Kansas City +31.3% into CPKC Stadium, San Diego +80.4% into Snapdragon, Seattle +98.8% into Lumen Field, Washington +145.0% into Audi Field, and Gotham +165.6% into Red Bull Arena (Figure 3). Washington's and Gotham's gains are the largest precisely because their 2021 move year, itself still COVID-suppressed, is excluded from the "after" average rather than counted as a depressed first season.

Fig. 3. Attendance trajectories for the five NWSL teams that relocated, with post-move seasons highlighted. Every move coincides with a large gain, but all five fall inside the league's 2021-2024 growth window.

Without season fixed effects the post-relocation indicator is worth +26.8% ($p = 0.347$), and with them it collapses to +5.8% ($p = 0.827$). All five moves fall inside the same window in which the league as a whole grew, so the indicator has almost no variation left once the league trend is absorbed. This reflects a data and design limitation, not that the relocation effect is insignificant. My sister project, *Gotham FC's Move to Queens: Does transit accessibility translate to attendance?* addresses this question with a synthetic control model to extract the effect of relocation from overall league growth.

4.6. A distinct media culture is forming outside the stadium.. Coverage volume has grown to 1,478 in 2024 (Figure 4). The four LDA topics separate cleanly. The largest, 3,471 articles, is broadcast and streaming coverage built around marquee fixtures, loading on *win*, *championship*, *rapino*, *uswnt*, and *megan*; that individual players' names carry this much weight across an 8,328-headline corpus is itself a measure of how much star power organizes attention. A second topic loads on *team*, *player*, *coach*, and *trade*, covering personnel news and the 2021 abuse-and-misconduct crisis. The remaining two cover schedules and power rankings, and general league and team news.

The composition of articles over time barely moves. Comparing 2013-2016 with 2021-2024, the broadcast-listing share falls from 46.3% to 40.5%, schedules and power rankings rise from 19.4% to 26.2%, coaching and controversy coverage falls from 28.4% to 23.6%, and general league news rises from 5.9% to 9.7%. The one clear compositional event is the 2021 crisis, when the coaching topic jumped to 29.7% from 23.1% two years earlier. Otherwise the league's coverage grew five times over without changing character. Peak analysis reinforces this. The broadcast-listing topic dominates 62.5% of peak months and 46.5% of peak days, so the largest spikes still track marquee events like championships and national-team tournaments.

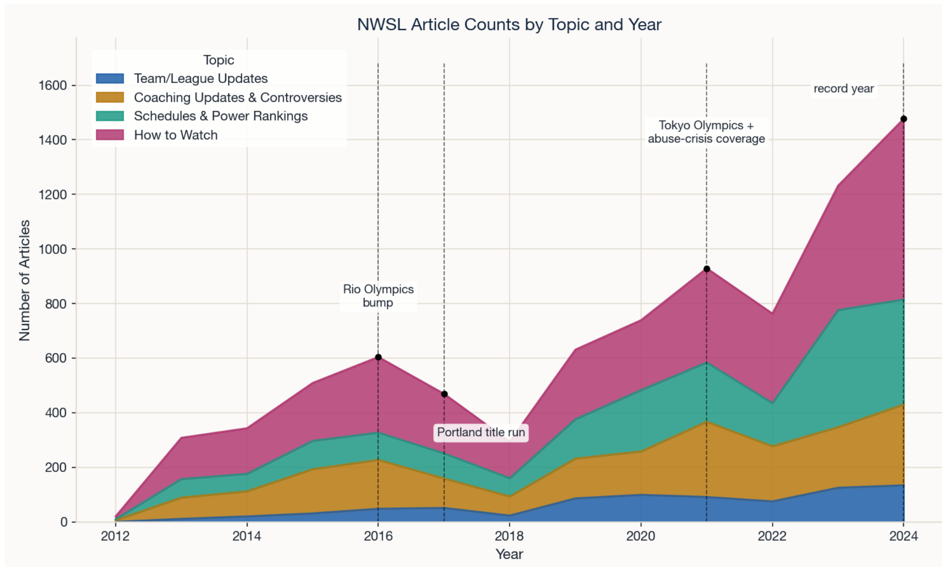

Fig. 4. LDA topic composition of 8,328 NWSL headlines by year, annotated with the real-world events behind the largest swings. Volume rises fivefold after 2018 while the topic mix stays remarkably stable, with the 2021 abuse-crisis coverage as the one clear compositional break.

Google search and YouTube data suggest the same thing. NWSL search interest rose from 6.1% of MLS's in 2018 to 17.4% in 2025. There has also been a rise of women's sports-specific media, which has far outgrown the league's own distribution. Just Women's Sports has 300,000 YouTube subscribers against the NWSL channel's 250,000, five independent women's-sports channels have launched since 2019, and their combined annual upload volume overtook the league's own channel in 2021 and has kept pulling away. The NWSL fan culture is being sustained by independent media, not just league marketing, a strong signal of durability.

5. Discussion

My first prediction that market size and on-field success are weak predictors of NWSL attendance holds for market size, but on-field success was a small predictor for both. My second prediction, that if that weakness reflects soccer rather than women's soccer, the same weakness appears in MLS holds. The final prediction, factors rooted in a club's own investment and fan culture, stadium relocation, proximity to downtown, and established rivalry, outperform the structural ones, holds for rivalries and downtown proximity, but the effects of stadium relocation are uncertain.

Read against the literature, these results favor the market-access account (1) over the enclosure account (2), since distance from downtown is negative in every specification; however, the estimated penalty is roughly a third of the published 6.6% per mile, which suggests the suburban penalty in the NWSL may be smaller than currently assumed, or that the earlier estimate absorbed variation this model assigns elsewhere.

The derby premium was strong in the NWSL, and 2.6 times stronger than in the MLS. Rivalries are built off of sustained cultural investment in fan relationships, unlike fixed factors like metro population. This signals the development of a women's sports-specific culture distinct from established men's leagues in the US. The media results are consistent with this finding: an audience that supports independent channels out-publishing the league is invested in the sport rather than in a marketing campaign.

These findings point more strongly to durable growth than a temporary fad. NWSL growth does not strongly depend on the conditions a front office cannot control, which is what makes it robust. This durability, however, must be qualified by the stratification of the league. The league is growing a whole, while the gap between its top and bottom widens, with Kansas City selling out a purpose-built stadium and Seattle filling under a fifth of a shared bowl. If relocation does carry a real effect, then the financial capacity to build a venue is precisely the kind of advantage that compounds.

5.1. Limitations. The estimates rest on 14 franchises and 1,005 games, with only 25 derbies and five relocations. The derby estimate is identified from two clubs in one metropolitan corridor, so it may reflect something specific to the Pacific Northwest rather than a general property of NWSL rivalries. Market size and downtown distance vary little within a team, so they are identified largely between teams and are vulnerable to any correlated team-level characteristic. The NWSL/MLS comparison drops distance and relocation because the downtown-coordinate lookup covers only NWSL markets, and all estimates here are correlational. The media analysis is bounded by MediaCloud's index and keyword filter, LDA topics are interpreted from term weights rather than hand-coded labels, and the six YouTube channels are a hand-selected sample.

5.2. Future work. The most direct extension is a causal design for the relocation question, isolating the effect of a new venue from the league trend it is currently confounded with; a companion project pursues this with synthetic control and transit-accessibility measures. The second is to connect the two halves of this paper by testing whether media growth in one season predicts attendance gains in the next, which would establish whether coverage leads demand or follows it. Both will become much more feasible as the league ages and stabilizes, since the NWSL is 13 years old, and this analysis rests on nine usable seasons.

6. Conclusion

The NWSL is not a smaller version of existing men's leagues; it is building its own distinct culture. Attendance in both the NWSL and MLS is largely unrelated to metro population and only mildly related to results, whereas derby fixtures were stronly correlated to attendance. The growth appears durable because it rests on a fan culture the league and its clubs are building, rather than on conditions they inherit. Whether the league distributes that growth evenly is a separate question, and the evidence here is less encouraging.

7. Agentic Analysis

Claude Code (Sonnet 5 medium-high) was used to assist this analysis. A detailed breakdown of what I accepted, what I rejected, and where it went wrong is available at this link .

8. References

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